

Grade Eight Science - Chemistry Assessment Checklist

Periodic Table and Chemical Bonding Information:

1. Know how to use the Periodic Table (location of metals, non-metals, and metalloids, atomic number (giving the number of electrons and protons), atomic mass, periods, groups, number of valence electrons, how to calculate the number of neutrons)
2. Know the difference between an element and a compound
3. Know how to draw the Bohr-Rutherford Diagram and electron dot diagrams for elements
4. Know what physical and chemical changes are
5. Know what physical (qualitative and quantitative) and chemical properties are
6. Know the difference between an ionic compound and a molecular compound
7. Know the difference between ionic and covalent bonding
8. Be able to draw bonding diagrams for ionic compounds (diagram with the arrows) and molecular compounds (sharing of electrons, electron dot diagram, and structural formula)
9. Be able to write names and chemical formulas for ionic compounds and molecular compounds

Interpret, Lab, and Data Based Information:

1. Know what independent (what you are changing to collect the data), dependent (what you are measuring - the data), and control variables (what is being kept the same throughout all trials of the experiment) are and be able to identify them
2. Know how to read and interpret data on a bar graph, data table, or paragraph and be able to draw conclusions from the data provided
3. Know how to write an experimental method (aka: procedure) in numbered, specific steps
4. Know how to identify experimental errors, outline the problem caused by them, and decide on a way to fix these errors for future experiments